

Generation Interconnection System Impact Study Report

For Queue Project Q15-0440

Metaline Falls 100 kV

January 25, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0440. The Interconnection Customer proposes a 104.1 MW solar photovoltaic facility interconnecting at the Metaline Falls 100 kV substation in the Ohio area. The request was received on February 22, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 189 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,632,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
GILCRET / METALINE FALLS 100 kV L	WELLS / FORT BRAGG 230 kV	158.7	127	100.0
GILCREST / WOBURN 100 kV	FT BRAGG / WELLS 230 kV	146.3	133	100.0
WOBURN / YORBA LINDA 100 kV L	FT BRAGG / WELLS 230 kV Line	117.5	127	77.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
METALINE FALLS / GILCREST SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$14,094,000	100% (\$14,094,000)
GILCREST / WOBURN 100 kV	Rebuild line with higher-capacity conductor	\$9,072,000	100% (\$9,072,000)
WOBURN / YORBA LINDA 100 kV line	Rebuild line with higher-capacity conductor	\$6,466,000	100% (\$6,466,000)
	Total allocated to project		\$29,632,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.